

Grounded Generation: Can an Epistemic Artifact Make LLM Exploration More Creative, Not Just More Honest?

Executive Summary

The present report provides an exhaustive evaluation of the generative layer within the physics-kernel shore-pack design. The program relies on an epistemic artifact featuring constraint-defined search spaces, characterized failure archives, quarantined speculation channels, and explicit gap definitions. While previous analyses have established the efficacy of this structure for policing epistemological honesty, the current inquiry addresses an entirely different and unproven dimension: whether this typed foundation inherently drives creative generation. Specifically, the analysis evaluates five core hypotheses (H11 through H15) regarding the generative power of constraints, the utility of characterized failures, the behavioral impact of consequence-free framing, the superiority of well-posed gaps, and the inevitable tension between verification pressure and generative diversity.

An exhaustive synthesis of the contemporary literature on automated scientific discovery, computational creativity, and large language model (LLM) alignment reveals a profound architectural tension between the physics-kernel paradigm and established successful discovery systems. Current state-of-the-art frameworks capable of generating verifiable technical novelty—such as FunSearch, AlphaEvolve, AlphaGeometry, and AlphaProof—operate in a high-compute, heavily sampled regime governed by flawless, deterministic ground-truth evaluators. In these architectures, the LLM functions strictly as a localized, highly stochastic mutation operator. Transformational creativity and novelty do not emerge zero-shot from the prompt scaffold; rather, they are emergent properties of the massive evolutionary search loop, island-based quality-diversity archiving, and the strict pruning mechanisms of formal execution environments.

Conversely, systems attempting to operate in a low-sampling, prompt-scaffolded regime without ground-truth execution—most notably the AI Scientist—exhibit severe epistemological breakdowns, characterized by hallucinated citations, template-filling, and the misclassification of standard baseline approaches as novel discoveries. The physics-kernel operates in a precarious middle ground: a low-budget, highly contextualized session environment checked by an admission gate that verifies structural coherence rather than objective physical truth.

Consequently, the empirical literature yields divided verdicts on the program's foundational hypotheses. The assertion that verification pressure collapses diversity (H15) is rigorously confirmed by the literature on typicality bias and reward-model overoptimization. The proposition that well-posed gaps outperform open exploration (H14) is similarly confirmed. However, the generative power of constraints (H11) and the utility of failure archives (H12) face significant headwinds in a low-sampling regime, primarily due to the "Accuracy-Correction Paradox," which demonstrates that strong LLMs frequently over-correct and degrade when fed their own localized failures without an external oracle. The value of sandboxed speculation (H13) remains unverified for technical novelty. Ultimately, the application of grounded creativity at the session scale is an open problem, requiring the program to recognize its status as an uncontrolled

frontier experiment and implement specific empirical tests to validate its architecture.

Question-by-Question Analysis

Q1: Automated Discovery Systems and the Locus of Novelty

Evaluating the physics-kernel design requires dissecting the architectures of modern automated discovery systems to isolate the exact locus of novelty. The central question is whether novelty arises from the prompt-side epistemic scaffolding (the instructions, constraints, and context) or from the computational search loop (sampling budget, diversity archives, and ground-truth evaluation).

The FunSearch architecture provides a definitive case study in combinatorial optimization and mathematical discovery. FunSearch achieved breakthroughs in the cap set problem and online bin packing by framing discovery as a search over a space of program fragments. The constraint structure is rigid: the system operates within a pre-defined program skeleton containing boilerplate code, requiring the LLM to evolve only a specific priority function. The failure memory structure relies on an island-based evolutionary algorithm; failed programs are simply discarded or retained only if they offer a structurally distinct, low-fitness seed to maintain diversity, while high-scoring programs are stored in a database to construct prompts for the next iteration. The sampling budget is immense, requiring millions of queries to the LLM. Crucially, the evaluator is a flawless, deterministic computational execution sandbox that scores the generated programs. Ablation studies and critical reviews locate the novelty entirely within the evolutionary database and the deterministic evaluator. The LLM acts merely as a localized mutation and crossover operator, possessing a shallow mathematical understanding. The discoveries emerge from the sheer scale of the search loop exploring the function space, not from the zero-shot creativity of the prompt. (Evidence: PEER-REVIEWED, NUMERICAL, FOLKLORE).

AlphaEvolve expands this paradigm to broader computational challenges, demonstrating the capacity to evolve entire code files rather than single functions. The constraint structure is defined by code execution boundaries and deterministic evaluation scripts that compile, run, and score the mutated candidates. Its failure memory utilizes a MAP-Elites-inspired Quality-Diversity archive to balance the exploitation of high-scoring algorithms with the exploration of novel, lower-scoring pathways. The sampling budget remains massive, executing hundreds of thousands of evaluations parallelized across supercomputer clusters. Similar to FunSearch, the locus of novelty resides in the autonomous improvement cycle and the evaluation cascade. The LLM provides the syntactic capacity to generate structured code diffs, but the discovery of functionally equivalent circuit simplifications or faster matrix multiplication algorithms is a byproduct of the evolutionary algorithm navigating the search space, driven by the absolute ground-truth signal of the compiler and execution environment. (Evidence: PEER-REVIEWED).

In the domain of formal mathematics, AlphaGeometry and AlphaProof represent the frontier of neuro-symbolic theorem proving. AlphaGeometry pairs a neural language model with a Deductive Database Arithmetic Reasoning (DDAR) symbolic engine, while AlphaProof utilizes the Lean 4 formal proof environment. The constraint structure is absolute: the system must produce proofs that perfectly satisfy the formal logic verifiers. The sampling budget relies on an initial pretraining phase using a staggering 21.8 million synthetic geometric theorems, followed by highly parallelized Monte Carlo Graph Search during inference. AlphaProof introduces a

failure memory mechanism via Test-Time Reinforcement Learning (TTRL), which generates millions of problem variants at inference time to adapt to specific bottlenecks. The locus of novelty is split between the massive synthetic pretraining distribution (which instills the pattern-matching required for auxiliary constructions) and the deterministic formal checker, which prunes infinite branching paths. The LLM does not reason its way to a novel proof zero-shot; it proposes thousands of intermediate steps that the symbolic engine mercilessly filters. (Evidence: PEER-REVIEWED).

The AI Scientist, developed by Sakana AI, offers a cautionary counterpoint, as it operates closer to the physics-kernel's low-budget, prompt-scaffolded regime. The system attempts to automate the entire research lifecycle using a low sampling budget (often under \$15 per paper) and standard LaTeX templates as constraint structures. Its evaluator relies heavily on LLM-as-a-judge mechanisms for peer review and basic compiler checks for code execution. Independent evaluations reveal a catastrophic failure to produce genuine technical novelty. The locus of novelty is effectively an illusion; the system relies on simplistic keyword searches, resulting in severe typicality bias where it classifies well-established, decades-old concepts (such as SGD micro-batching) as novel discoveries. Furthermore, it exhibits critical failure rates in execution (42% coding error failure) and produces manuscripts riddled with hallucinated citations (18% to 55% fabrication rates depending on the underlying model). The AI Scientist demonstrates that without a robust ground-truth evaluator and a massive sampling budget, reliance on prompt-side instructions and LLM self-evaluation leads to template-filling and simulated discovery rather than authentic epistemic advancement. (Evidence: PREPRINT, NUMERICAL).

Finally, literature-based discovery (LBD) systems, such as AGATHA, operate on fundamentally different architectures. Tracing back to Swanson's A-B-C model, these systems identify hidden connections within existing scientific corpora (e.g., if A relates to B, and B relates to C, an unexplored connection may exist between A and C). AGATHA utilizes a massive static semantic graph comprising billions of edges extracted from historical texts, embedding these elements to learn a data-driven ranking criterion. The evaluator is temporal holdout validation, predicting post-2015 connections using pre-2015 data. The locus of novelty lies in the mathematical topology of the graph embeddings and semantic proximity. Creativity in LBD is defined strictly as combinatorial retrieval and link prediction, which does not translate to the generative construction of novel physical formalisms required by the physics-kernel. (Evidence: PEER-REVIEWED).

The overwhelming consensus of the architectural literature indicates that novelty is a property of the search loop and the objective evaluator, not the prompt-side scaffold.

Q2: Constraint and Creativity

This section evaluates H11 (constraints generate) and H14 (well-posed gaps beat open invitations). The inquiry spans theoretical definitions of computational creativity, human psychology, and modern techniques in constrained language generation.

In computational creativity research, Margaret Boden's foundational theory distinguishes between combinatorial, exploratory, and transformational creativity. Exploratory creativity involves navigating an existing conceptual space defined by a set of generative rules and acceptability criteria. Transformational creativity, which yields the most profound scientific paradigm shifts, occurs when the underlying rules or axioms defining the conceptual space are deliberately altered, discarded, or transcended. Formalized through directed acyclic graphs, transformational creativity requires precise modifications to the "enabling constraints" of a

domain. Translated to LLM inference, constraints are the mathematical boundaries of the conceptual space. Open-ended invitations fail because they provide no topological boundaries, causing the model to collapse into the center of its pre-trained probability distribution. A well-posed gap, which explicitly defines the boundaries and the required closing conditions, establishes the exact shape of the conceptual space, enabling systematic exploration. (Evidence: PEER-REVIEWED; Transfer to LLM: INFERENCE).

The psychological literature further supports the generative utility of constraints. Patricia Stokes' "creative constraints" framework demonstrates that extreme freedom frequently results in reliance on highly accessible, familiar solutions. The Combinatorial Theory of Constraints posits that creativity requires paired constraints: one dimension that precludes reliance on existing reliable paths (forcing divergent thinking) and another that promotes novel criteria. When applied to human subjects, these constraints force cognitive processes to reroute, discovering stepping stones that a completely unconstrained search would bypass. By explicitly stating constraints affirmatively—outlining the shape of the admissible region rather than providing an exhaustive list of prohibitions—a system focuses cognitive (or computational) attention on emergent problem-solving. (Evidence: HUMAN, PEER-REVIEWED; Transfer to LLM: INFERENCE).

In the specific context of LLM inference, constrained text generation studies provide a computational parallel to psychological findings. Unconstrained autoregressive generation naturally gravitates toward high-probability token sequences, leading to generic outputs. Frameworks like Constrained Discrete Diffusion (CDD) demonstrate that imposing strict constraints directly into the sampling process alters the probability landscape, treating valid programs or texts as energy minima within a constraint-shaped landscape. This forces the model to generate tokens that satisfy complex logic rules or structural requirements without sacrificing semantic coherence or fluency. Furthermore, prompt-side ideation experiments reveal that structured, multi-constraint instructions—especially when presented in an optimized "hard-to-easy" order—yield higher novelty and conceptual depth than freeform brainstorming. The constraints effectively act as a negative prompt against the model's base training distribution, preventing semantic drift and forcing the attention mechanism to synthesize concepts along the boundaries of the defined gap. (Evidence: PEER-REVIEWED, HEURISTIC).

Verdicts:

- **H11 (constraints generate): PLAUSIBLE.** Known rigorously in human psychology and formal computational theory; known heuristically in prompt engineering. Explicitly defining the boundaries of the conceptual space forces the generation landscape out of high-probability generic distributions.
- **H14 (well-posed gaps beat open invitations): CONFIRMED.** Known rigorously. Open invitations consistently result in typicality bias and convergence toward the mean. Providing explicit closing conditions defines the topological search space necessary for targeted exploration.

Q3: Failure Archives as Generative Material

This section evaluates H12 (wreckage is fuel), examining whether presenting characterized failures and precise margins of failure improves subsequent proposals, moving beyond simple prohibition adherence.

In evolutionary computation, the utility of low-fitness archives is well-established through Quality-Diversity (QD) algorithms. In highly complex, deceptive landscapes, optimization algorithms that greedily follow gradients toward an objective function frequently become trapped

in local optima. Algorithms such as MAP-Elites abandon strict objective optimization in favor of maintaining a multidimensional archive of phenotypically diverse solutions. Crucially, this archive includes "failed" regions or low-fitness individuals that exhibit high behavioral novelty. These low-fitness solutions act as vital "stepping stones," allowing the evolutionary search to traverse deceptive valleys and discover novel basins of attraction that would be entirely inaccessible to a purely objective-driven search. (Evidence: PEER-REVIEWED).

In classical Artificial Intelligence, case-based reasoning and near-miss learning provide a theoretical foundation for utilizing characterized failures. Patrick Winston's early work demonstrated that providing a learning system with a "near miss"—an example that lacks exactly one essential feature of the target concept—accelerates learning significantly faster than providing purely positive examples. Contrasting a positive instance with a structurally similar failure highlights the precise boundaries of concept membership. When translated to LLM exploration, an exclusion reservoir that provides the exact margins and mechanisms of a killed approach functions as a highly dense semantic near-miss, offering gradient-like directional information about the boundary of the admissible hypothesis space. (Evidence: PEER-REVIEWED; Transfer to LLM: INFERENCE).

However, the efficacy of feeding failure margins back into LLMs is heavily complicated by recent findings in the self-correction literature. Frameworks like Reflexion demonstrate that models can utilize verbal reinforcement and episodic memory to adjust action sequences. Yet, extensive ablations reveal a severe limitation termed the "Accuracy-Correction Paradox." Intrinsic self-correction—where a model attempts to fix its own errors without an external ground-truth oracle—is largely ineffective. The paradox dictates that weaker models can correct simple, shallow errors, but highly capable frontier models generate "deep" errors that resist intrinsic correction. Furthermore, providing explicit error location hints or self-generated critiques frequently degrades performance, causing the model to over-correct and alter an initially correct reasoning step into an incorrect one. Because the physics-kernel operates without a ground-truth simulator, feeding highly detailed failure margins back into the LLM prompt risks triggering this over-correction pathology, leading to degeneration loops rather than productive stepping stones. (Evidence: PREPRINT, NUMERICAL).

Verdict for H12 (wreckage is fuel): PLAUSIBLE but highly contingent (Structural). Known rigorously in evolutionary computation (stepping stones) and classical AI (near-misses). However, due to the Accuracy-Correction Paradox, transferring this to a prompt-scaffolded LLM session without an external oracle is risky. The model may lack the internal reasoning depth to utilize the failure margin constructively, instead oscillating into over-correction.

Q4: Sandboxed Speculation and Risk-Taking

This section evaluates H13 (quarantine buys boldness), which posits that designating a typed, non-contaminating channel for ungrounded conjecture actively increases productive risk-taking and exploration distance.

The premise relies on manipulating the model's contextual framing. Literature demonstrates that LLMs are profoundly susceptible to instruction framing effects. Altering variables such as outcome severity, temporal urgency, or procedural justice in a prompt can shift an LLM's ethical and strategic recommendations by dramatic margins—up to 27 to 54 percentage points. By assigning specific personas or framing a task within a distinct context, users can significantly alter the model's tone, hedging behavior, and compliance levels. (Evidence: PREPRINT, PEER-REVIEWED).

Models aligned via Reinforcement Learning from Human Feedback (RLHF) develop deeply

embedded safety guardrails and typicality biases, resulting in pervasive sycophancy and excessive hedging when operating in uncertain domains. These models are trained to avoid definitive statements on unverified topics to prevent hallucination penalties. Consequently, instructing a model that it is operating in a "draft," "brainstorming," or "quarantined speculation" mode theoretically alters the active semantic manifold. By explicitly removing the penalty for incorrectness, the framing reduces the activation of safety and refusal vectors, allowing the model to sample from lower-probability, higher-variance regions of its pre-trained knowledge base without triggering alignment guardrails. (Evidence: PREPRINT, HEURISTIC). Architecturally, the principle of decoupling unconstrained generation from strict verification is validated by speculative decoding in inference acceleration. In this paradigm, a small draft model rapidly proposes sequences of tokens at zero penalty, which are subsequently verified in parallel by a strict target model. Tokens that fail alignment are discarded without polluting the final output. While this is a hardware-level acceleration technique rather than an epistemological one, the underlying architecture confirms that separating consequence-free drafting from rigorous gating allows for high-variance exploration without compromising the integrity of the final verified state.

Verdict for H13 (quarantine buys boldness): UNVERIFIED (Local/Structural). Known heuristically that framing alters tone and reduces hedging. However, there is no rigorous empirical literature demonstrating that framing a channel as consequence-free explicitly yields *higher quality or more novel technical physics conjectures*. The exact transfer of stylistic boldness into scientific utility remains an unstudied gap.

Q5: Diversity Collapse Under Verification Pressure

This section quantifies H15 (the verifier taxes diversity), examining the established phenomenon of mode collapse under alignment and optimization pressure.

The empirical literature provides overwhelming confirmation that verification pressure, reward modeling, and alignment protocols (such as RLHF and DPO) systematically decimate generative diversity. This phenomenon, widely documented as mode collapse, occurs when an aligned model collapses its output distribution onto an exceedingly narrow, high-reward subset of modes, effectively abandoning the vast semantic diversity it acquired during pretraining. The root cause lies in the Kullback-Leibler (KL) divergence regularizer used in preference learning. Theoretical analyses via social choice theory prove that the KL penalty heavily biases the model toward majority preferences, causing the probability of generating a majority-preferred output to far exceed the actual population preference, resulting in severe reductions in per-token entropy and lexical diversity. (Evidence: PEER-REVIEWED, NUMERICAL).

This collapse is exacerbated by "typicality bias." Human preference data inherently favors fluent, familiar, and predictable text. When optimization algorithms train against these biased reward models, the system suffers from "Reward Model Overoptimization" (an manifestation of Goodhart's Law), where the proxy reward score increases but genuine output quality, robustness, and diversity plateau or degrade. Best-of-N sampling and filtering mechanisms identically narrow the distribution, discarding tail-end probabilities that often harbor novel, transformational concepts. (Evidence: PEER-REVIEWED).

To counteract diversity collapse, recent research proposes explicit interventions. "Soft Preference Learning" (SPL) decouples the entropy and cross-entropy terms within the KL penalty, allowing developers to independently regulate generation diversity and restore proportional representation of concepts without inducing the severe grammatical degradation caused by crude temperature scaling. At the inference level, "Verbalized Sampling" (VS)

provides a training-free mitigation. By instructing the LLM to generate a distribution of diverse responses alongside their estimated probabilities, the prompt forces the model to bypass the single-mode trap, frequently recovering up to 66% of the base model's lost diversity and boosting variance by 1.6x to 2.1x in creative tasks. Furthermore, Quality-Diversity algorithms dictate that diversity must be maintained as an explicit constraint (via separate dimensional archives) rather than hoped for as a byproduct of sampling.

Verdict for H15 (the verifier taxes diversity): CONFIRMED. Known rigorously. Any optimization, filtering, or strict gating mechanism will inevitably induce mode collapse and entropy reduction. Grounded-exploration systems require explicit, mathematically enforced diversity preservation machinery to offset this fundamental algorithmic tax.

3. The Novelty Locus Table (Q1)

System	Constraint Structure	Failure Memory	Sampling Budget	Evaluator Type	Where the Evidence Locates the Novelty
FunSearch	Pre-defined program skeleton; evolves specific priority functions only.	Island-based database; low-scoring variants discarded or retained strictly to seed diversity.	Millions of LLM queries.	Flawless, deterministic ground-truth code execution (combinatorial).	Search Loop & Evaluator: Ablations prove the LLM is merely a localized mutation operator. Novelty stems from evolutionary scaling and absolute computational verification.
AlphaEvolve	Full code file evolution bounded by deterministic API contracts.	MAP-Elites / QD-style program database balancing exploration and exploitation.	Hundreds of thousands, parallelized across accelerator clusters.	Deterministic simulators, compilers, and supercomputer environments.	QD Archive & Scale: Cross-model adversarial pressure and diversity maintenance traverse complex algorithmic pathways; the LLM merely translates intent to syntax.
AI Scientist	Standardized	Iterative	Extremely low	LLM-as-a-judge	No genuine

System	Constraint Structure	Failure Memory	Sampling Budget	Evaluator Type	Where the Evidence Locates the Novelty
(Sakana)	LaTeX templates and predefined GitHub codebases.	execution with runtime limits; high failure to escape local bugs.	(session-scale; <\$15 per paper generation).	peer review; basic Python execution checks.	novelty: Heavy template filling, high hallucination rates (18-55%), and misclassification of standard baselines due to typicality bias and lack of a ground-truth oracle.
AlphaGeometry / AlphaProof	Strict formal proof languages (Lean 4) and DDAR engines.	Test-Time RL generates millions of problem variants at inference to bypass bottlenecks.	Millions of queries; pretrained on 21.8M synthetic geometries.	Unforgiving symbolic deduction engine and formal kernel verifiers.	Formal Checker & Pretraining Data: The deterministic evaluator relentlessly prunes the infinite branching space. Pretraining provides the pattern matching for auxiliary constructions.
AGATHA (LBD)	Semantic A-B-C relational graph embeddings over scientific corpora.	Static embedding ranking; no generative failure feedback loop.	Static inference over a pre-computed graph of billions of edges.	Temporal holdout validation (predicting future literature links).	Graph Topology: Novelty is defined entirely as identifying latent, high-probability connections in existing text embeddings, not generating new formalisms.

4. Feasibility Table for Q6: The Generative Pack Assessment

Step	Best Supporting Evidence	Best Contrary Evidence	Main Gap / Hidden Assumption	Verdict & Earliest Fatal Step
1. State well-posed gaps (H14)	Clear instruction boundaries reduce hallucination; LBD systems rely on precisely targeted A-C concept gaps to focus retrieval.	Completely unconstrained novelty search sometimes yields serendipitous discoveries that rigid framing precludes.	Assumes the defined gap is mathematically accurate and computationally tractable for the LLM's specific latent space.	PLAUSIBLE
2. Affirmative constraint geometry (H11)	Constrained Discrete Diffusion and human psychology studies show strict limitations force divergent, novel problem-solving.	Overly narrow constraint formulation restricts the search space, potentially preventing true transformational creativity.	Assumes textual constraints in a prompt effectively shape LLM latent probability distributions equivalently to rigid mathematical boundaries.	PLAUSIBLE
3. Present characterized wreckage (H12)	Near-miss learning theory; MAP-Elites stepping stones bypass deceptive local minima traps.	The Accuracy-Correction Paradox: providing precise error feedback to strong models often triggers severe over-correction and degradation without an external oracle.	Assumes the LLM possesses the intrinsic reasoning depth to utilize the failure margin constructively rather than hallucinating a flawed workaround to avoid the prohibited tokens.	NEEDS REFINEMENT (Earliest Fatal Step)
4. Two-channel quarantine (H13)	Framing effects shift LLM tone and reduce hedging significantly; speculative decoding benefits from decoupled draft phases.	RLHF safety weights are deeply embedded; altering prompt framing cannot entirely bypass pre-trained typicality bias and mode collapse.	Assumes "zero cost" framing actually alters the model's internal statistical generation priorities toward valid technical novelty, rather than just stylistic looseness.	UNVERIFIED
5. Preserve	RLHF/DPO	Maintaining vast	Assumes disjoint	CONFIRMED

Step	Best Supporting Evidence	Best Contrary Evidence	Main Gap / Hidden Assumption	Verdict & Earliest Fatal Step
diversity deliberately (H15)	demonstrably cause mode collapse; QD archives and Verbalized Sampling are strictly required to maintain search diversity.	QD archives mathematically requires high sampling budgets, directly conflicting with the low-session architecture.	seeds and prompt engineering are sufficient to overcome entropy collapse in a low-sampling regime.	(Mitigation is structurally mandated)

Earliest Fatal Step Identified: Step 3 (Present characterized wreckage). This step is highly vulnerable due to the Accuracy-Correction Paradox. Providing precise failure mechanisms to a frontier LLM in a low-budget session without an immediate, ground-truth simulator risks inducing over-correction. The model is statistically likely to abandon sound foundational reasoning simply to avoid the specifically highlighted error tokens, resulting in a degenerative loop rather than a productive stepping stone.

5. Critical Obstructions Identified

The fundamental obstruction to the physics-kernel generative layer is an irreconcilable architectural mismatch between the program's operating regime and the established mechanics of automated discovery. This manifests as the **"Search Loop vs. Prompt Scaffold" Crux**. The prompt author explicitly raises this crux: *If novelty comes from massive sampling and a perfect evaluator, H11-H14 might be true but will not transfer to a low-sampling, prompt-scaffolded session.* The literature exhaustively validates this fear. Verified discovery systems (FunSearch, AlphaEvolve, AlphaGeometry) achieve novelty by treating the LLM as a highly stochastic, error-prone mutation operator. They rely on the brute-force execution of millions of generated paths against an absolute, deterministic ground truth (a compiler, simulator, or formal Lean/DDAR verifier). Creativity, in these systems, is an emergent property of the **search loop's massive scale** and the **quality-diversity archive's breadth**, not the semantic elegance of the prompt.

The physics-kernel attempts to replicate this outcome in a fundamentally opposite environment: a handful of long, highly-contextualized sessions, checked by a deterministic admission gate that verifies *structural coherence* (adherence to the claim graph) rather than *ground truth* (objective physical reality). This creates two fatal structural bottlenecks:

1. **The Sampling Deficit and Mode Collapse:** Without millions of queries, the LLM will inevitably fall victim to typicality bias. Constrained by RLHF, the model will output the most probabilistically average, familiar response. While disjoint seeds (Step 5) mitigate this slightly, prompt-based diversity generation cannot mathematically replace the expansive exploration of a MAP-Elites search space traversing deceptive landscapes.
2. **The Oracle Deficit and The Correction Paradox:** Because the gate checks structural formatting rather than empirical truth, an LLM generating ungrounded physical speculation (even in the quarantine channel) cannot receive the immediate, objective binary feedback required to fuel a true evolutionary loop. It must rely on its intrinsic reasoning capabilities to iterate. The literature on the "Accuracy-Correction Paradox" proves this is deeply flawed; frontier models routinely degrade and over-correct when

attempting to fix their own deep logical errors without external ground-truth validation. Therefore, while applying constraints (H11) and failure margins (H12) via prompt scaffolding will shape the output format flawlessly, it will not spontaneously instill the capacity for zero-shot transformational creativity.

6. Bottom-Line Assessment

Single Verdict: NEEDS REFINEMENT.

The generative-pack design is structurally elegant and leverages valid theoretical frameworks spanning cognitive psychology, near-miss learning, and diversity preservation. However, its foundational assumption—that prompt-side epistemic scaffolding can effectively substitute for massive-scale computational evaluation in generating technical scientific novelty—is entirely unsupported by the consensus of modern automated discovery literature. Grounded creativity at the session scale is currently an uncontrolled frontier experiment.

Strongest Defensible Reformulation: The generative layer must be recast. It cannot function as a zero-shot discovery engine; it must operate as an **Evolutionary Proposal Generator** that explicitly acknowledges the LLM as a vulnerable mutation operator subject to typicality bias.

1. Retain the well-posed gaps (H14) and explicit constraint geometry (H11) to rigorously define the topological boundaries of the search space.
2. Retain the explicit diversity preservation (H15), but formalize it algorithmically. Implement "Verbalized Sampling"—instructing the model to generate multiple distinct hypotheses with self-assigned probabilities—to force the attention mechanism out of the single-mode typicality trap.
3. Downgrade expectations for the failure archive (H12). Use characterized failures as negative constraints, but avoid providing hyper-detailed logical refutations, as this risks triggering the Accuracy-Correction Paradox and subsequent degenerative loops.

The Minimal Informative In-Program Experiment (Stage 6): Because session-scale grounded creativity is unstudied in the literature, the program must empirically test H12 and H13 internally to establish baseline metrics.

- *Test 1 (Failure Fuel / H12):* Execute parallel sessions targeting an identical gap. Group A receives the exclusion reservoir as a bare list of prohibited approaches. Group B receives the exact failure mechanisms and margins. **Metric:** Measure the survival rate through the structural gate and the semantic novelty of the generated claims. If Group B exhibits lower survival, the Accuracy-Correction Paradox is active.
- *Test 2 (Quarantine Boldness / H13):* Execute parallel sessions. Group A is instructed that all outputs must pass the rigorous admission gate. Group B is explicitly granted the "quarantine channel" and instructed to maximize risk-taking. **Metric:** Measure the semantic embedding distance (variance) of the proposed hypotheses from the base graph to quantify whether consequence-free framing objectively unlocks transformational creativity or merely alters stylistic tone.

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